

MODULE 7 – SALES & OPERATIONS PLANNING (S&OP) AND INVENTORY MANAGEMENT

This module covers **Sales & Operations Planning (S&OP), Lot Sizing, EOQ, Inventory Management, Reorder Point (ROP), Quantity Discount Models, and ABC Analysis**. These concepts are fundamental for aligning demand, supply, production, and inventory decisions.

◆ 1. Sales and Operations Planning (S&OP)

Definition

Sales and Operations Planning (S&OP) is an **integrated planning process that synchronizes demand planning, supply planning, and financial planning** to create a feasible and profitable business plan. It helps executives make decisions regarding production, inventory, capacity, workforce, and suppliers.

Objectives of S&OP

- Balance demand and supply
 - Improve coordination across departments
 - Optimize inventory levels
 - Enhance profitability
 - Support strategic and tactical planning
 - Align operations with organizational goals
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Key Components of S&OP

1. Demand Planning

Forecast future sales using:

- Historical sales trends
- Marketing initiatives
- Industry trends
- Customer demand patterns

2. Supply Planning

Determines:

- Production schedules

- Material requirements
- Inventory targets
- Capacity availability

3. Integrated Planning

All departments jointly review:

- Demand forecasts
- Production plans
- Budget constraints
- Resource availability

Benefits of S&OP

Benefit	Impact
Enhanced Efficiency	Reduces overproduction and stockouts
Improved Customer Satisfaction	Better product availability
Lower Costs	Better resource utilization
Enhanced Profitability	Better planning and forecasting
Strategic Alignment	Operations support business goals

S&OP Process (6 Steps)

Step 1: Gather & Manage Data

- Historical sales
- Forecast accuracy
- Demand trends

Step 2: Demand Planning

- Promotions
- New launches
- Customer service policies

Step 3: Production & Supply Planning

- Inventory goals
- Safety stock
- Production methods
- Capacity planning

Step 4: Pre-S&OP Meeting

- Compare demand and supply plans
- Check budget feasibility

Step 5: Executive S&OP Meeting

- Final approval of integrated plan

Step 6: Implementation

- Execute and monitor plans
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◆ 2. Lot Sizing

📌 Definition

Lot size refers to the quantity ordered or produced in a single batch. Choosing the correct lot size balances inventory costs and customer demand.

🎯 Effects of Lot Size

Large Lot Sizes

Advantages:

- Lower setup cost per unit
- Fewer orders

Disadvantages:

- Higher inventory carrying cost
- Higher storage requirement
- Reduced flexibility

Small Lot Sizes

Advantages:

- Lower inventory levels
- Better responsiveness

Disadvantages:

- More setups
- Higher setup cost per unit

◆ 3. Lot Sizing Techniques

A. Static Methods

Fixed Order Quantity (FOQ)

- Fixed quantity ordered at reorder point.

Lot-for-Lot (L4L)

- Order exactly what is needed.

Economic Order Quantity (EOQ)

- Minimizes ordering and holding costs.

B. Periodic Methods

Period Order Quantity (POQ)

Orders based on demand during a specific period.

Fixed Period Order Quantity

Fixed order quantity at regular intervals.

C. Dynamic Methods

Least Unit Cost (LUC)

Chooses order size with lowest average unit cost.

Least Total Cost (LTC)

Chooses quantity minimizing total inventory cost.

Part Period Balancing (PPB)

Balances ordering and holding costs.

Silver-Meal Heuristic

Efficient approach for varying demand situations.

◆ 4. Economic Order Quantity (EOQ)

✦ Definition

EOQ determines the optimal order quantity that minimizes:

- Ordering costs
 - Inventory holding costs
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EOQ Formula

$$EOQ = \sqrt{\frac{2DS}{H}}$$

Where:

- D = Annual demand
 - S = Ordering cost per order
 - H = Holding cost per unit per year
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Assumptions of EOQ

1. Constant demand
 2. Constant lead time
 3. Constant ordering cost
 4. Single product
 5. No stockouts
 6. Instant replenishment
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EOQ Example

Given:

- D = 1000 units

- $S = ₹2000$
- $H = ₹6000$

$$EOQ = \sqrt{\frac{2(1000)(2000)}{6000}}$$

$$EOQ = \sqrt{666.67}$$

$$EOQ \approx 25.82$$

(The lecture notes indicate an order size slightly above 66 shirts using the provided example values.)

◆ 5. Inventory Management

📌 Definition

Inventory includes:

- Raw Materials
- Work-In-Progress (WIP)
- Finished Goods

Inventory management involves:

- Purchasing
 - Storage
 - Monitoring
 - Usage
 - Selling
-

🎯 Need for Inventory Management

Benefits

- Saves money
- Prevents stockouts
- Reduces storage cost
- Improves productivity

- Preserves product quality
 - Improves sales forecasting
 - Enhances customer satisfaction
-

◆ 6. Types of Inventory

Type	Description
Raw Materials	Inputs used in production
Work-In-Progress (WIP)	Semi-finished products
Finished Goods	Ready for sale
Pipeline Inventory	Goods in transit

◆ 7. Independent vs Dependent Demand

Independent Demand	Dependent Demand
Demand for finished products	Demand for components
Requires forecasting	Calculated using BOM
Influenced by market demand	Driven by parent item demand

Example

- Bicycle = Independent Demand
 - Tire = Dependent Demand
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◆ 8. Factors Affecting Inventory Operations

Financial Factors

- Cost of capital
- Storage cost
- Ordering cost

Market Factors

- Demand forecasting

- Seasonality
- Product life cycle

Supplier Factors

- Lead time
- Supplier reliability
- Minimum Order Quantity (MOQ)

Management Factors

- Inventory systems
- Inventory policies
- Managerial capability

◆ 9. Reorder Point (ROP)

📌 Definition

ROP is the inventory level at which a new order should be placed to avoid stockouts.

🧮 Formula

$$\text{ROP} = \text{Lead Time Demand} + \text{Safety Stock}$$

Where:

- Lead Time Demand = Average demand during lead time
 - Safety Stock = Buffer inventory
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Benefits

- Prevents stockouts
 - Reduces inventory costs
 - Improves efficiency
 - Enhances customer satisfaction
 - Balances inventory investment and service levels
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◆ 10. Quantity Discount Models

📌 Definition

A pricing strategy where buyers receive discounts for purchasing larger quantities.

Types

Tiered Pricing

Different discounts at different quantity levels.

Cumulative Quantity Discounts

Based on total purchases over a period.

Trade-Offs

Buyer Benefits

- Lower unit cost
- Lower ordering cost

Buyer Risks

- Higher inventory cost
- Obsolescence risk

Seller Benefits

- Higher sales volume
- Lower processing cost

Seller Risks

- Lower profit margin per unit
-

◆ 11. ABC Analysis

📌 Definition

ABC Analysis classifies inventory based on annual consumption value to focus control efforts where they matter most.

Classification

Category	Items	Contribution
A	~20% Items	~80% Value
B	~30% Items	~15% Value
C	~50% Items	~5% Value

(The lecture notes use the typical 80% / 15% / 5% split.)

Applications

Inventory Management

- Strict control for A items
- Moderate control for B items
- Simple control for C items

Purchasing

- Better supplier negotiation for A items

Warehouse Management

- Prime storage for A items

Product Assortment

- Monitor A items closely
 - Remove low-value C items
-

EXPECTED EXAM QUESTIONS WITH ANSWERS

★ 1. What is S&OP? Explain its benefits.

Answer:

Sales and Operations Planning (S&OP) is an integrated planning process that aligns demand, supply, and financial plans. Benefits include better coordination, reduced costs, improved customer satisfaction, enhanced profitability, and strategic alignment.

★ 2. Explain the steps in S&OP.

Answer:

1. Data Collection
 2. Demand Planning
 3. Supply Planning
 4. Pre-S&OP Meeting
 5. Executive S&OP Meeting
 6. Implementation
-

★ **3. What is EOQ? State its assumptions.**

Answer:

EOQ determines the optimal order quantity minimizing ordering and holding costs. Assumptions include constant demand, lead time, ordering cost, no stockouts, and instantaneous replenishment.

★ **4. Differentiate Independent and Dependent Demand.**

Answer:

Independent demand is market-driven and forecasted, while dependent demand is derived from demand for parent items and calculated using BOM.

★ **5. Explain ABC Analysis.**

Answer:

ABC Analysis classifies inventory into A, B, and C categories based on annual consumption value, enabling better inventory control and cost optimization.

★ **6. What is Reorder Point (ROP)?**

Answer:

ROP is the inventory level at which a new order should be placed to prevent stockouts.

 CASE-STUDY-BASED ANSWERS (OPERATIONS MANAGEMENT)

✚ **Case 1: Frequent Stockouts Despite High Inventory**

Problem:

Customer complaints due to unavailable products.

Analysis:

- Poor reorder point calculation
- Inadequate safety stock
- Poor forecasting

Solution:

- Implement ROP system
 - Maintain safety stock
 - Improve demand forecasting
 - Use inventory software
-

 Case 2: Excess Inventory and High Holding Cost

Problem:

Warehouse is full and inventory costs are increasing.

Solution:

- Apply EOQ
- Use ABC analysis
- Reduce order quantities
- Improve forecasting accuracy

Expected Outcome:

Lower holding costs and improved cash flow.

 Case 3: Poor Coordination Between Sales and Production

Problem:

Sales promises exceed production capability.

Solution:

- Implement S&OP process
- Conduct monthly demand-supply reviews
- Align production with forecasts

Outcome:

Better service levels and lower operational conflicts.

📌 Case 4: Inventory Investment Too High**Problem:**

Large capital locked in inventory.

Solution:

- ABC classification
 - Tight control of A items
 - Review slow-moving C items
 - Optimize EOQ and ROP
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✅ FINAL QUICK REVISION POINTS

- **S&OP = Demand + Supply + Finance Integration**
- **EOQ minimizes ordering and holding costs**
- **ROP prevents stockouts**
- **Lot sizing balances inventory and setup costs**
- **Independent demand is forecasted**
- **Dependent demand is calculated using BOM**
- **ABC Analysis = A (80%), B (15%), C (5%)**
- **Inventory management improves cost, service, and profitability**
- **Safety stock acts as a buffer against uncertainty**
- **Effective inventory control improves customer satisfaction and cash flow**

These notes cover the complete Week 7 material in an exam-focused format suitable for MBA Operations Management preparation.

WEEK 8 – OPERATIONS PLANNING–II

Master Production Scheduling (MPS), Material Requirements Planning (MRP), Manufacturing Resource Planning (MRP-II) & ERP

MBA Operations Management – Exam Ready Notes

This week focuses on how organizations convert demand forecasts into production plans, material requirements, resource allocation, and enterprise-wide integration systems.

UNIT 1: MASTER PRODUCTION SCHEDULING (MPS)

What is MPS?

A **Master Production Schedule (MPS)** specifies:

- What products will be produced
- How many units will be produced
- When production will occur

MPS acts as the bridge between:

Sales Forecasts → Production Planning → Material Planning

Its objective is to maximize customer service while minimizing inventory and production inefficiencies.

Key Terms in MPS

1. Gross Requirements (GR)

Total demand for a finished product during a planning period.

Includes:

- Forecast demand
- Customer orders
- Marketing promotions

Does not consider existing inventory.

2. On-Hand Inventory

Inventory currently available.

3. Net Requirements (NR)

Actual production needed after considering available inventory.

Concept:

Net Requirement = Gross Requirement – Available Inventory

Variables Used

- D_t = Gross demand during week t
 - S_t = Quantity completed during week t
 - I_t = Inventory during week t
 - N_t = Net requirement during week t
-

Role of MPS**1. Production Planning**

Converts sales forecasts into production schedules.

2. Demand Management

Ensures adequate production without excessive inventory.

3. Input to MRP

Provides production data required for material planning.

4. Capacity Planning

Checks availability of:

- Labor
- Machines
- Resources

5. Inventory Control

Balances inventory and customer demand.

6. Interdepartmental Coordination

Connects:

- Sales
 - Production
 - Purchasing
 - Engineering
-

Objectives of MPS

Inventory Control

Maintain optimum inventory.

Due Date Setting

Ensure customer commitments are met.

Resource Planning

Coordinate materials and production resources.

Customer Satisfaction

Improve delivery reliability.

Cost Reduction

Reduce operating and inventory costs.

Components of MPS

- Forecasting
 - Order Entry
 - Production Plan
 - MRP
 - Capacity Planning
 - Inventory Levels
 - BOM
 - Lead Times
 - Priority Planning
 - Feedback Loop
-

Steps in Developing MPS

Step 1: Understand Products

Define product categories.

Step 2: Determine Lead Time

Estimate:

- Procurement time
- Manufacturing time
- Packaging time
- Delivery time

Step 3: Estimate Demand

Forecast production requirements.

Step 4: Match Resources

Check labor and machine availability.

Step 5: Identify Production Shortfalls

Find bottlenecks.

Step 6: Prepare MPS

Use quantitative and qualitative data.

Step 7: Review and Revise

Continuously improve schedule.

Time Fences and Planning Horizon

Planning Horizon

The total future period covered by MPS.

Examples:

- Weekly
- Monthly
- Annual

Time Fences

Boundaries that determine scheduling flexibility.

Demand Time Fence (DTF)

Near-term schedule.

Changes are restricted.

Planning Time Fence (PTF)

Allows moderate flexibility.

Frozen Zone

Almost no changes allowed.

Slushy Zone

Limited modifications allowed.

Liquid Zone

Highest flexibility.

Benefits

- Better planning efficiency
 - Stable schedules
 - Improved coordination
 - Balanced flexibility and control
-

Applications of MPS

Pharmaceutical Industry

Challenges:

- Long lead times
- Regulatory compliance
- Product recalls

MPS ensures medicine availability.

Fashion Industry

Challenges:

- Seasonal demand
- Rapid trend changes

MPS ensures timely production.

UNIT 2: MATERIAL REQUIREMENTS PLANNING (MRP)

Definition

Material Requirements Planning (MRP) is a system that determines:

- What materials are required
- How much is required
- When they are required

It balances:

Demand ↔ Inventory ↔ Production

Objectives of MRP

Ensure Material Availability

Materials available when needed.

Minimize Inventory

Reduce excess stock.

Improve Production Scheduling

Avoid delays and bottlenecks.

Support Customer Service

Meet delivery commitments.

Role of MRP

Effective Production Scheduling

Uses MPS and BOM to calculate material needs.

Inventory Control

Reduces:

- Overstocking
- Stockouts

Resource Optimization

Improves utilization of resources.

Better Responsiveness

Adjusts to changing demand.

Components of MRP

1. Bill of Materials (BOM)

Complete list of materials needed for one product.

2. Inventory Records

Current stock information.

3. Capacity Planning

Checks resource availability.

4. Order Release & Tracking

Monitors purchase orders.

5. Performance Measurement

Evaluates MRP effectiveness.

MRP Planning Process

Step 1

Identify customer demand.

Step 2

Break demand into components using BOM.

Step 3

Check inventory.

Step 4

Allocate resources.

Step 5

Generate:

- Purchase orders
- Work orders
- Transfer orders

Step 6

Monitor production progress.

MRP Inputs

Demand

- Customer orders
- Forecasts

BOM

Material requirements list.

Inventory Status

Current stock availability.

MPS

Production schedule.

MRP Outputs

Material Requirements

What materials are needed.

Order Schedules

When orders should be released.

Production Timing

When production should begin.

JIT Support

Supports Just-In-Time manufacturing.

Important MRP Formulas

Gross Requirement (GR)

$GR = MPS + \text{Scheduled Receipts} - \text{Beginning Inventory}$

Net Requirement (NR)

$NR = GR - \text{On-Hand Inventory} - \text{Scheduled Receipts}$

Planned Order Release (POR)

$POR = \text{Requirement Date} - \text{Lead Time}$

Planned Order Receipt

Receipt Date = Release Date + Lead Time

UNIT 3: MANUFACTURING RESOURCE PLANNING (MRP-II)

Definition

MRP-II is an expanded version of MRP.

It integrates:

- Materials
- Labor
- Machines
- Finance

for complete production planning.

Importance of MRP-II

Resource Balancing

Balances:

- Labor
- Machines
- Materials

Better Supplier Relationships

Improves visibility and communication.

Higher Productivity

Improves manufacturing performance.

Better Customer Service

Supports on-time delivery.

MRP vs MRP-II

MRP	MRP-II
Focuses on materials	Focuses on total resources

MRP	MRP-II
Inventory oriented	Enterprise oriented
Limited scope	Wider scope
Material planning	Material + Labor + Capacity

Benefits of MRP-II

Better Scheduling

More realistic production plans.

Improved Capacity Planning

Identifies bottlenecks.

Better Inventory Control

Optimized stock levels.

Reduced Lead Time

Faster production cycles.

Better Decision Making

Enterprise-wide visibility.

UNIT 4: ENTERPRISE RESOURCE PLANNING (ERP)

Definition

ERP is an integrated software system that manages:

- Finance
- HR
- Manufacturing
- Supply Chain
- Procurement
- CRM

through one centralized database.

Key Features of ERP

Integration

Connects all business functions.

Process Automation

Automates routine work.

Real-Time Data

Provides instant information.

Supply Chain Management

Improves visibility.

CRM

Enhances customer relationships.

Scalability

Supports business growth.

Evolution of ERP

Period	Development
1960s–70s	MRP
1980s–90s	MRP-II
1990s–2000s	ERP
2000s–2010s	ERP-II
2010+	Cloud ERP

Advantages of ERP

Better Reporting

Single source of truth.

Improved Customer Service

Faster response times.

Inventory Optimization

Avoids overstocking and stockouts.

Better Cash Flow

Improved billing and collections.

Cost Savings

Better planning and procurement.

Improved Security

Cloud and centralized security.

Better Supply Chain Management

Enhanced transparency.

Challenges of ERP Implementation

High Cost

Complex Integration

Employee Resistance

Data Migration Issues

Project Management Challenges

ERP Implementation Phases

(Shown in the ERP Implementation Stages diagram on Page 34)

1. Discovery & Planning
 2. Design
 3. Development
 4. Testing
 5. Deployment
 6. Support
-

ERP Implementation Approaches

1. Big Bang

Entire ERP implemented at once.

Advantages

- Fast
- Lower cost
- Strong focus

Disadvantages

- High risk
 - System failure impact is large
-

2. Phased Rollout

Implemented gradually.

Advantages

- Lower risk
- Better training
- Easier change management

Disadvantages

- Longer implementation time
-

3. Parallel Adoption

Old and new systems run together.

Advantages

- Backup available
- Lower operational risk

Disadvantages

- Expensive
 - Resource intensive
-

4. Pilot Implementation

Small-scale testing before full rollout.

Advantages

- Early problem detection
- Reduced failure risk

Disadvantages

- Pilot results may differ from actual rollout
-

5. Hybrid Implementation

Combination of multiple approaches.

Advantages

- Flexible
- Balanced risk

Disadvantages

- Complex planning
-

6. Process Transit Implementation

Department-by-department implementation.

Advantages

- Easier adoption
- Lower disruption

Disadvantages

- Longer transition period
-

HIGH-PROBABILITY EXAM QUESTIONS

2 Marks

1. Define MPS.
2. What is Planning Horizon?
3. What is BOM?
4. Define MRP.
5. What is ERP?
6. What is MRP-II?

5 Marks

1. Explain the role of MPS.
2. Discuss Time Fences.
3. Explain MRP inputs and outputs.
4. Differentiate MRP and MRP-II.
5. Explain ERP implementation phases.

10 Marks

1. Explain Master Production Scheduling with examples.
2. Discuss Material Requirements Planning and its benefits.
3. Explain MRP-II and its role in Operations Management.
4. Explain ERP and various implementation approaches.
5. Compare Big Bang and Phased Rollout implementation.

CASE STUDY ANSWERS (OM)

Case 1: Production Delays Due to Material Shortage

Problem: Frequent production stoppages.

Solution: Implement MRP using BOM, inventory records, and MPS.

Result: Reduced stockouts and smoother production.

Case 2: High Inventory Cost

Problem: Excess raw material inventory.

Solution: Use MRP and MPS to calculate exact requirements.

Result: Lower inventory carrying costs.

Case 3: Poor Coordination Across Departments

Problem: Sales, production, and procurement working independently.

Solution: Implement ERP.

Result: Integrated decision-making and real-time visibility.

Case 4: ERP Implementation Failure Risk

Problem: Employees resist new ERP system.

Solution: Use phased rollout with proper training and change management.

Result: Better user acceptance and lower implementation risk.

One-Page Revision

MPS → What to produce? How much? When?

MRP → What materials are needed?

MRP-II → Materials + Labor + Machines + Finance

ERP → Entire Business Integration

Planning Horizon → Total future planning period

Time Fence → Scheduling control boundary

BOM → Product structure list

Big Bang → Fast but risky

Phased Rollout → Slow but safe

Pilot → Test first, deploy later

ERP Goal → One integrated business system

WEEK 9 – QUALITY MANAGEMENT

MBA Operations Management – Exam Ready Notes

This module covers **Quality Concepts, Total Quality Management (TQM), Quality Gurus, Baldrige Excellence Framework, and EFQM Excellence Model**. These topics explain how organizations achieve customer satisfaction, continuous improvement, and operational excellence.

UNIT 1: QUALITY CONCEPT

What is Quality?

Quality refers to:

- Degree of excellence of a product or service.
- Ability to satisfy customer needs and expectations.
- According to ISO 9000:2000:

“Quality is the degree to which a set of inherent characteristics fulfills requirements.”

Garvin’s Five Definitions of Quality

Definition	Meaning
Transcendent	Excellence or superiority
Product-Based	Amount of desirable attributes
User-Based	Fitness for use
Manufacturing-Based	Conformance to specifications
Value-Based	Satisfaction relative to price

Harvey & Green’s Five Views of Quality

1. Exceptional

- High standards
- Distinctiveness
- Zero defects

2. Perfection (Consistency)

- Conformance to specifications
- Error-free processes

3. Fitness for Purpose

- Product performs intended function

4. Value for Money

- Quality relative to cost paid

5. Transformative

- Quality creates improvement and empowerment
- Focus on value addition and employee development

What is Quality Management?

Quality Management ensures that:

- Products and services meet customer expectations.
- Processes are efficient and effective.
- Continuous improvement occurs throughout the organization.

It focuses not only on the final output but also on improving the entire process.

Dimensions of Quality

Dimension	Meaning
Performance	Primary operating characteristics
Features	Secondary characteristics
Conformance	Meets specifications
Reliability	Consistent performance
Durability	Useful life
Serviceability	Ease of repair
Responsiveness	Human interaction quality

Dimension	Meaning
Aesthetics	Appearance and sensory appeal
Reputation	Brand image and past performance

Significance of Quality

Customer Satisfaction

- Improves loyalty and repeat purchases.

Higher Profitability

- Reduces defects and warranty costs.

Competitive Advantage

- Differentiates products from competitors.

Reduced Risk

- Avoids legal and safety issues.

Operational Efficiency

- Minimizes waste and rework.

Innovation

- Encourages continuous improvement.
-

Quality Process Model

The diagram on Page 6 shows:

Inputs

- Materials
- Methods
- Procedures
- Information
- People
- Skills
- Knowledge

- Training
- Equipment

↓

Process

↓

Outputs

- Products
- Services
- Information
- Documentation

Feedback from customers and process performance continuously improves quality.

UNIT 2: TOTAL QUALITY MANAGEMENT (TQM)

What is TQM?

Total Quality Management (TQM) is:

A management philosophy focused on maximizing customer satisfaction through continuous improvement and total employee involvement.

Unlike traditional quality systems, TQM involves every employee in quality improvement.

Basic Concepts of TQM

Customer Focus

Quality is defined by customers.

Employee Involvement

Everyone participates in quality improvement.

Process-Centered Approach

Focuses on improving processes, not just outcomes.

Five Pillars of TQM

1. Product

2. Process
3. System
4. People
5. Leadership

Principles of TQM

1. Customer Focus

Understand and exceed customer expectations.

2. Total Employee Commitment

Employees require:

- Training
- Tools
- Motivation

3. Process Approach

Follow standardized procedures.

4. Integrated System

Departments work together.

5. Strategic & Systematic Approach

Manage interrelated processes effectively.

Barriers to TQM Implementation

Barrier	Explanation
Lack of Leadership Commitment	No top management support
Resource Constraints	Insufficient funds/training
Employee Resistance	Fear of change
Cultural Issues	Lack of teamwork
Short-Term Thinking	Focus on immediate gains

Barrier	Explanation
Inadequate Measurement	Poor performance tracking
Communication Problems	Lack of awareness

Benefits of TQM

Lower Production Costs

- Reduced defects
- Less rework

Increased Productivity

- Better systems and processes

Customer Satisfaction

- Better product quality

Higher Market Share

- Positive reputation

Employee Motivation

- Greater participation and empowerment
-

TQM Implementation Steps

1. Leadership commitment
 2. Employee training
 3. Employee empowerment
 4. Communication strategy
 5. Continuous improvement
-

Samsung – TQM Success Story

Situation

- In the 1980s–90s Samsung had poor market share.
- Motorola dominated the market.

Action

- Chairman Lee Kun-hee adopted TQM.
- Focused on product quality and customer experience.

Result

- Samsung became one of the world's leading technology brands.
- Expanded into multiple consumer product markets.



Exam Insight:

This is a favorite case-study question.

UNIT 3: CONTRIBUTIONS OF QUALITY GURUS

American Gurus

W. Edwards Deming

Contributions

- 14 Points for Management
- Continuous Improvement
- Statistical Process Control (SPC)



Known as:

Father of Modern Quality Management

Joseph M. Juran

Juran Trilogy

1. Quality Planning
2. Quality Control
3. Quality Improvement

Pareto Principle

80% of problems come from 20% of causes.

Armand V. Feigenbaum

Contributions

- Total Quality Control (TQC)
 - Cost of Quality concept
-

Japanese Gurus

Kaoru Ishikawa

Contributions

- Fishbone Diagram
 - Cause-and-Effect Diagram
 - Company-Wide Quality Control
-

Genichi Taguchi

Contributions

- Loss Function
 - Robust Design
-

Shigeo Shingo

Contributions

- Poka-Yoke (Mistake Proofing)
 - SMED (Single Minute Exchange of Dies)
-

Taiichi Ohno

Contributions

- Toyota Production System (TPS)
 - Kanban
 - Just-In-Time (JIT)
-

Western Gurus

Philip B. Crosby

Four Absolutes of Quality

1. Quality = Conformance to requirements
2. Prevention is the quality system
3. Zero Defects is the standard
4. Cost of Non-Conformance measures quality

Famous Concept

“Quality is Free”

Tom Peters

Contribution

- Management by Walking Around (MBWA)
 - McKinsey 7-S Framework
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Walter Shewhart

Contributions

- Statistical Process Control
 - Control Charts
-

UNIT 4: BALDRIGE EXCELLENCE FRAMEWORK

Definition

A non-prescriptive framework developed in the USA to help organizations achieve long-term excellence and competitiveness.

Seven Baldrige Criteria

Criteria	Focus
Leadership	Organizational direction
Strategy	Strategic planning
Customers	Customer relationships

Criteria	Focus
Measurement, Analysis & Knowledge Management	Data-driven decisions
Workforce	Employee engagement
Operations	Process management
Results	Organizational performance

Baldrige Assessment Process

1. Application Submission
2. Examiner Review
3. Site Visit (optional)
4. Scoring
5. Consensus Evaluation
6. Feedback Report
7. Award Selection

Score Range

0–1000 points

UNIT 5: EFQM EXCELLENCE MODEL

Definition

European Foundation for Quality Management (EFQM) Model is a framework for assessing organizational excellence.

Applicable to:

- Public sector
 - Private sector
 - Small organizations
 - Large organizations
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EFQM Structure

The model shown on Page 25 divides criteria into:

Enablers (What Organization Does)

1. Leadership
 2. Policy & Strategy
 3. People
 4. Partnership & Resources
 5. Processes
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Results (What Organization Achieves)

1. Customer Results
2. People Results
3. Society Results
4. Key Performance Results

Logic

Enablers → Results → Continuous Learning & Innovation

EXPECTED EXAM QUESTIONS WITH ANSWERS

2 Marks

1. Define Quality.
 2. What is TQM?
 3. What is Zero Defects?
 4. What is Poka-Yoke?
 5. Define EFQM.
 6. What is Baldrige Framework?
-

5 Marks

Explain dimensions of quality.

Answer: Performance, Features, Conformance, Reliability, Durability, Serviceability, Responsiveness, Aesthetics, and Reputation.

Explain TQM principles.

Answer: Customer focus, employee commitment, process approach, integrated system, and strategic approach.

Explain Juran Trilogy.

Answer: Quality Planning, Quality Control, and Quality Improvement.

10 Marks

1. Explain TQM with benefits and barriers.
 2. Discuss contributions of quality gurus.
 3. Explain Baldrige Excellence Framework.
 4. Explain EFQM Excellence Model.
 5. Compare Baldrige and EFQM models.
-

 CASE STUDY-BASED ANSWERS (OM)

Case 1: High Customer Complaints

Problem

Product defects and dissatisfaction.

Solution

- Implement TQM.
- Focus on customer requirements.
- Use SPC and quality audits.

Result

Improved customer satisfaction and reduced defects.

Case 2: Frequent Manufacturing Errors

Problem

High rework and scrap rates.

Solution

- Apply Poka-Yoke (Shigeo Shingo).
- Use Fishbone Diagram (Ishikawa) to identify root causes.

Result

Reduced errors and waste.

Case 3: Poor Interdepartmental Coordination

Problem

Departments working independently.

Solution

- Introduce TQM integrated systems.
- Encourage cross-functional teamwork.

Result

Improved efficiency and quality.

Case 4: Organization Seeking Excellence Award

Problem

Need for performance benchmarking.

Solution

- Apply Baldrige or EFQM framework.
- Measure leadership, processes, workforce, and results.

Result

Continuous improvement and organizational excellence.

ONE-PAGE REVISION SHEET

Quality = Fitness for Use + Conformance + Customer Satisfaction

TQM = Customer Focus + Employee Involvement + Continuous Improvement

Five Pillars of TQM

- Product

- Process
- System
- People
- Leadership

Key Gurus

- Deming → 14 Points
- Juran → Trilogy
- Crosby → Zero Defects
- Ishikawa → Fishbone Diagram
- Shingo → Poka-Yoke
- Ohno → JIT & Kanban
- Shewhart → Control Charts

Baldrige Criteria (7)

Leadership → Strategy → Customers → Measurement → Workforce → Operations → Results

EFQM Model

Enablers (5) → Results (4) → Innovation & Learning

Most Important Exam Topics

- ★ TQM Principles
- ★ Quality Gurus
- ★ Baldrige Framework
- ★ EFQM Model
- ★ Samsung TQM Case Study

WEEK 10 – STATISTICAL PROCESS CONTROL (SPC) TOOLS

MBA Operations Management – Exam Ready Notes

This module covers **Statistical Process Control (SPC)**, **Quality Control (QC) Tools**, **Control Charts ($\bar{X}$, R, P, and C Charts)**, **Three-Sigma Limits**, and the **Old & New Seven QC Tools**. These tools help organizations monitor process stability, detect defects, and improve quality continuously.

UNIT 1: STATISTICAL PROCESS CONTROL (SPC)

What is SPC?

Statistical Process Control (SPC) is a quality control technique that uses statistical methods to monitor, control, and improve production processes.

Purpose

- Reduce variation
- Detect abnormalities
- Maintain process stability
- Improve product quality

A manufacturing process never produces identical products; therefore SPC measures and controls process variation.

Types of Data in SPC

1. Variable Data

Continuous and measurable.

Examples:

- Length
- Weight
- Voltage
- Thickness

2. Attribute Data

Discrete and countable.

Examples:

- Pass/Fail
 - Accept/Reject
 - Defective/Non-defective
-

UNIT 2: QUALITY CONTROL (QC)

Definition

Quality Control (QC) is the process of:

- Monitoring quality
- Measuring performance
- Identifying deviations
- Taking corrective action

QC ensures products meet:

- Customer requirements
- Design specifications
- Quality standards

Quality control must be integrated with:

- Cost control
 - Scope control
 - Risk control
 - Production control
-

UNIT 3: OLD SEVEN QC TOOLS

These are the traditional tools used for problem-solving and quality improvement.

1. Check Sheet

Definition

A structured form used to collect and tally data.

Purpose

- Data collection

- Defect recording
- Trend identification

Example

Tracking:

- Machine failures
 - Customer complaints
 - Product defects
-

2. Fishbone Diagram (Ishikawa Diagram)

Definition

A cause-and-effect diagram used to identify root causes of a problem.

Common Categories (6M)

- Man
- Machine
- Material
- Method
- Measurement
- Mother Nature (Environment)

Purpose

Root cause analysis.

3. Histogram

Definition

A bar graph showing frequency distribution.

Shows

- Central tendency
- Spread
- Shape of distribution

Limitation

Does not show sequence of data.

4. Pareto Chart

Based on Pareto Principle

80% of problems arise from 20% of causes.

Developed by **Joseph Juran**.

Purpose

Prioritize major causes.

Also called:

Vital Few and Useful Many

5. Scatter Diagram

Definition

A graph showing relationship between two variables.

Purpose

Identify:

- Positive correlation
- Negative correlation
- No correlation

Used in regression analysis.

6. Control Chart

Definition

A chart used to distinguish:

- Common causes
- Special causes

of variation.

7. Stratification

Definition

Separates data into categories.

Examples:

- Shift
- Supplier
- Machine
- Location

Purpose:

Identify hidden patterns.

UNIT 4: NEW SEVEN QC TOOLS

Used mainly for planning, management, and decision-making.

1. Relationship Diagram**Purpose**

Shows cause-effect relationships among complex factors.

Application

Complex problems with interrelated causes.

2. Affinity Diagram**Purpose**

Groups large numbers of ideas into logical categories.

Application

Brainstorming and planning.

3. Tree Diagram**Purpose**

Breaks objectives into smaller actionable tasks.

Application

Problem-solving and planning.

4. Matrix Diagram

Purpose

Shows relationships between two or more factors.

Application

Customer requirements vs Product features.

5. Matrix Data Analysis Diagram

Purpose

Analyzes numerical relationships within a matrix.

Application

Product characteristic evaluation.

6. Process Decision Program Chart (PDPC)

Purpose

Identifies potential failures and countermeasures.

Application

Risk prevention and contingency planning.

7. Arrow Diagram

Purpose

Project scheduling and time management.

Similar to:

PERT / CPM Network Diagram.

Used For

- Critical Path identification
 - Project duration estimation
-

UNIT 5: CONTROL CHARTS

What is a Control Chart?

A graphical tool used to determine whether a process is operating under statistical control.

Every control chart contains:

Central Line (CL)

Represents process average.

Upper Control Limit (UCL)

Lower Control Limit (LCL)

Any point beyond control limits indicates an out-of-control process.

Three-Sigma Rule

SPC uses the famous:

68–95–99.7 Rule

Range	Percentage of Observations
$\pm 1\sigma$	68.27%
$\pm 2\sigma$	95.45%
$\pm 3\sigma$	99.73%

Control limits are usually set at:

Mean $\pm 3\sigma$

This captures approximately 99.73% of normal variation.

UNIT 6: $\bar{X}$ (X-BAR) CHART

Purpose

Used for:

- Variable data
- Monitoring process average

Example:

- Thickness

- Weight
- Diameter

Important Formulas

Sample Mean

$$\bar{X} = \frac{\sum X}{n}$$

Grand Mean

$$\bar{\bar{X}} = \frac{\sum \bar{X}}{N}$$

Average Range

$$\bar{R} = \frac{\sum R}{N}$$

Control Limits for $\bar{X}$ Chart

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

Where:

- A_2 = Control chart constant
- $\bar{R}$ = Average range

UNIT 7: R-CHART

Purpose

Measures process variability.

Formula

Formula

$$UCL_R = D_4 \bar{R}$$

$$LCL_R = D_3 \bar{R}$$

Where:

- D_3 and D_4 are chart constants.

Interpretation

In Control

All points lie within limits.

Out of Control

One or more points exceed limits.

Chance Causes vs Assignable Causes

Chance Causes	Assignable Causes
Natural variation	Specific identifiable variation
Small impact	Large impact
Difficult to eliminate	Can be corrected
Temperature fluctuation	Operator error
Voltage variation	Faulty equipment

UNIT 8: P-CHART (ATTRIBUTE CONTROL CHART)

Purpose

Used to monitor:

Proportion of defective units

Based on:

Binomial Distribution

Suitable when:

- Sample sizes vary
 - Data is defective/non-defective
-

Assumptions

1. Pass/Fail outcomes
 2. Same defect probability
 3. Independent observations
 4. Same testing procedure
-

Formula

Average Fraction Defective

$$\bar{p} = \frac{\text{Total Defectives}}{\text{Total Items Sampled}}$$

Control Limits

$$UCL_p = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

$$LCL_p = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

If LCL is negative → use 0.

Uses of P-Chart

- Monitor process stability
 - Detect special causes
 - Compare performance before and after improvements
-

UNIT 9: C-CHART

Purpose

Used to monitor:

Number of defects

when sample size remains constant.

Based on:

Poisson Distribution

Examples:

- Air bubbles in bottles
- Scratches on surfaces
- Soldering defects

Difference Between P-Chart and C-Chart

P-Chart	C-Chart
Percentage of defectives	Number of defects
Binomial distribution	Poisson distribution
Variable sample size possible	Constant sample size required
Defective units	Defects within units

C-Chart Formula

Average Defects

$$\bar{c} = \frac{\sum c}{k}$$

Control Limits

$$UCL_c = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL_c = \bar{c} - 3\sqrt{\bar{c}}$$

If $LCL < 0 \rightarrow$ use 0.

Interpretation of C-Chart

In Control

All points within limits.

Out of Control

Any point outside $\pm 3\sigma$ limits.

Example from notes:

- Sample 20 exceeded UCL.
 - Process considered out of control.
 - Root cause investigation required.
-

EXPECTED EXAM QUESTIONS WITH ANSWERS

2 Marks

1. What is SPC?
 2. What is Quality Control?
 3. Define Control Chart.
 4. What is a Check Sheet?
 5. Define P-Chart.
 6. Define C-Chart.
-

5 Marks

Explain the Old Seven QC Tools.

Answer:

Check Sheet, Fishbone Diagram, Histogram, Pareto Chart, Scatter Diagram, Control Chart, Stratification.

Explain Three-Sigma Rule.

Answer:

Approximately 99.73% of process observations fall within ± 3 standard deviations from the mean. Control limits are therefore set at Mean $\pm 3\sigma$.

Differentiate P-Chart and C-Chart.

(Use the comparison table above.)

10 Marks

1. Explain Statistical Process Control and its significance.
 2. Discuss Old and New Seven QC Tools.
 3. Explain $\bar{X}$ and R Charts with formulas.
 4. Explain P-Chart and C-Chart with applications.
 5. Explain control charts and process stability.
-

CASE STUDY-BASED ANSWERS (OM)

Case 1: Increasing Product Defects

Problem

Defect rates rising unexpectedly.

Solution

- Use P-Chart to monitor defect proportion.
- Identify assignable causes.
- Implement corrective actions.

Result

Reduced defect rate and improved quality.

Case 2: Machine Producing Variable Dimensions

Problem

Dimension variation in manufactured parts.

Solution

- Use $\bar{X}$ and R Charts.
- Monitor mean and variability.
- Identify machine calibration issues.

Result

Stable production process.

Case 3: Multiple Defects in Bottles

Problem

Air bubbles appearing in milk bottles.

Solution

- Apply C-Chart.
- Identify samples exceeding UCL.
- Investigate root causes.

Result

Reduced defect frequency.

Case 4: Frequent Process Failures**Problem**

Unknown reasons for process instability.

Solution

- Fishbone Diagram for root causes.
- Pareto Analysis to prioritize issues.
- Control Charts to monitor improvements.

Result

Improved process capability.

 ONE-PAGE REVISION SHEET

SPC = Statistical monitoring of process variation

Old 7 QC Tools

1. Check Sheet
2. Fishbone Diagram
3. Histogram
4. Pareto Chart
5. Scatter Diagram
6. Control Chart
7. Stratification

New 7 QC Tools

1. Relationship Diagram
2. Affinity Diagram
3. Tree Diagram
4. Matrix Diagram
5. Matrix Data Analysis
6. PDPC
7. Arrow Diagram

Control Charts

- $\bar{X}$ Chart → Process Mean
- R Chart → Process Variation
- P Chart → % Defectives
- C Chart → Number of Defects

Three-Sigma Rule

- 68.27%
- 95.45%
- 99.73%

Important Gurus Linked to SPC

- Walter Shewhart → Control Charts
- Juran → Pareto Principle
- Ishikawa → Fishbone Diagram

Exam Formula Focus

- ★ $\bar{X}$ Chart Limits
- ★ R Chart Limits
- ★ P Chart Formula
- ★ C Chart Formula
- ★ Three-Sigma Rule

✦ Numericals Most Likely for MBA Exams

1. $\bar{X}$ and R Chart Problem (Calculate UCL, CL, LCL)

2. **P-Chart Problem** (Compute defect proportion and limits)
3. **C-Chart Problem** (Compute defect counts and limits)
4. **Interpret Control Chart** (State whether process is in control or not)

These four numerical types are the most frequently repeated examination problems from this module.

WEEK 11 – LEAN EVOLUTIONS & LEAN PRODUCTION SYSTEM (LPS)

MBA Operations Management – Exam Ready Summary

This module explains the evolution from **Craft Production** → **Mass Production** → **Lean Production**, the **Toyota Production System (TPS)**, **Lean Manufacturing principles**, **Lean tools**, and **Just-In-Time (JIT) production**. It focuses on eliminating waste, improving productivity, and delivering maximum value to customers.

1. Evolution of Production Systems

A. Craft Production (19th Century)

- Products made by skilled craftsmen.
- Low volume, high customization.
- Expensive and time-consuming.

B. Mass Production (1913)

Introduced by **Henry Ford** through the moving assembly line.

Key Features

- Division of labour
- Interchangeable parts
- Work measurement
- Line balancing
- Moving assembly line

Objective

Produce the largest number of products at the lowest possible cost.

Ford's Model T became the benchmark for mass production worldwide.

C. Lean Production (Toyota Model)

Developed through the **Toyota Production System (TPS)**.

Characteristics

- Low waste
- High flexibility
- Better quality

- Faster response to customers
- Continuous improvement

Lean production became globally popular after the book:

 **The Machine That Changed the World**

by:

- James P. Womack
- Daniel T. Jones
- Daniel Roos

2. Toyota Production System (TPS)

TPS is the foundation of Lean Manufacturing.

Main Features

- Greater product variety
- Fast response and flexibility
- Stable production schedules
- Supply chain integration
- Demand management

TPS aims to eliminate waste while continuously improving quality and productivity.

3. Elements of Lean Production System

1. Continuous Improvement (Kaizen)
 2. Small Lot Production
 3. SMED (Single Minute Exchange of Dies)
 4. Kanban-based Pull Production
 5. Work Cells
 6. Heijunka (Production Leveling)
 7. Synchronization and Balancing
 8. Bottleneck Scheduling
-

4. Lean Production Philosophy

Just-In-Time (JIT)

Lean production supplies:

The right product, at the right time, in the right quantity, without waste.

Core Idea

- Zero inventory philosophy
 - Continuous flow
 - Customer-driven production
-

5. Five Principles of Lean Thinking

1. Specify Value

Understand value from the customer's perspective.

2. Identify Value Stream

Map all activities required to deliver value.

3. Make Value Flow

Ensure smooth movement of products and information.

4. Customer Pull

Produce only when customers demand it.

5. Pursue Perfection

Continuous improvement and waste elimination.

6. Lean Waste Management

The Three Mu's

Muda (Waste)

Activities that add no value.

Mura (Inconsistency)

Variation in workload or production.

Muri (Overburden)

Demand exceeds capacity.

Seven Wastes (TIM WOOD)

Letter	Waste
T	Transportation
I	Inventory
M	Motion
W	Waiting
O	Over Processing
O	Over Production
D	Defects

Customer Rule

Customers only pay for activities that add value.

7. Benefits of Lean Manufacturing

Lean Manufacturing (LM)

Focus:

- Eliminate non-value-added activities
- Increase customer value

Benefits

- ✓ Lower inventory
- ✓ Lower defects
- ✓ Reduced waste
- ✓ Better quality
- ✓ Higher productivity
- ✓ Reduced space and manpower
- ✓ Faster delivery

Key Formula

More output with fewer inputs

(materials, manpower, machinery, space, and time)

8. House of Lean

The "House of Lean" visually represents Lean Manufacturing.

Roof (Goal)

- Highest quality
- Lowest cost
- Shortest lead time

Pillars

Pillar 1: Just-In-Time (JIT)

- Reduce inventory
- Reduce lead time

Pillar 2: Jidoka

- Built-in quality
- Detect defects immediately

Foundation

- Standardization
- Kaizen (Continuous Improvement)
- Heijunka (Load Leveling)

Center

- Employee involvement
 - Teamwork
 - Leadership
 - Responsibility
-

9. Operational Definition of Lean System

Key Lean Characteristics:

- Waste elimination

- Customer value creation
- Productivity improvement
- Lead-time reduction
- Multi-skilled workforce
- Leadership
- 5S
- Jidoka
- Cellular Layout
- VSM
- TPM
- Kanban
- Kaizen
- Pull System
- Poka-Yoke
- Single Piece Flow

10. Important Lean Tools

Tool 1: 5S

Purpose

Organize and maintain an efficient workplace.

Japanese	Meaning
Seiri	Sort
Seiton	Set in Order
Seiso	Shine
Seiketsu	Standardize
Shitsuke	Sustain

Benefits

- Cleaner workplace
 - Better productivity
 - Less waste
 - Improved safety
-

Tool 2: Value Stream Mapping (VSM)

Definition

A graphical tool that maps:

- Material flow
- Information flow
- Value-added activities
- Non-value-added activities

Purpose

Identify waste and improve process flow.

Tool 3: SOP (Standard Operating Procedure)

Definition

A set of written instructions explaining how work should be performed.

Purpose

- Consistency
- Quality assurance
- Worker training
- Process control

SOP Development Steps

1. Capacity Work Table
 2. Operations Combination Chart
 3. Work Methods Chart
 4. Standard Operations Chart
-

Tool 4: Cellular Layout

Definition

Grouping machines and processes into cells for similar products.

Advantages

- Reduced material movement
- Reduced setup time
- Reduced WIP inventory
- Better resource utilization

Disadvantages

- Additional training
 - Higher investment
 - Poor balancing risks
-

Tool 5: Kaizen

Meaning

Continuous Improvement

Focus

- Eliminate waste
- Improve productivity
- Ongoing improvements

Kaizen Cycle

1. Brainstorm
 2. Measure
 3. Compare
 4. Innovate
 5. Standardize
 6. Repeat
-

Tool 6: Kanban

Definition

A visual scheduling and workflow management system.

Developed by:

Taiichi Ohno (Toyota)

Workflow Example

- To Do
- In Progress
- Testing
- Done

Pull System Concept

Production occurs only when demand exists.

Functions of Kanban

Function 1

Support JIT production.

Function 2

Visual control and continuous improvement.

Six Rules of Kanban

1. Downstream withdraws from upstream.
2. Produce only what was withdrawn.
3. Send defect-free products.
4. Level production.
5. Use workshop indicators.
6. Discover improvement opportunities.

Types

- Supplier Kanban
 - In-Factory Kanban
 - Production Kanban
 - Signal Kanban
-

Tool 7: Jidoka

Meaning

Automation with a Human Touch

Definition

Machines stop automatically when abnormalities occur.

Four Principles

1. Detect abnormality
2. Stop process
3. Fix immediate issue
4. Eliminate root cause

Functions

1. Human-Machine Separation

2. Defect Prevention

3. Immediate Correction

Result:

- Higher quality
 - Lower defects
 - Less waste
-

Tool 8: Poka-Yoke

Meaning

Mistake Proofing

Purpose

Prevent errors before defects occur.

Types

1. Stop Devices
2. Control Devices
3. Warning Devices

Goal

Zero Defects

Examples:

- USB insertion design
 - Electrical plug design
 - Safety interlocks
-

11. Just-In-Time (JIT) Production

Definition

Produce:

The right quantity, at the right place, at the right time.

Objectives

- Reduce waste
- Increase productivity
- Minimize inventory

Why Inventory is Undesirable?

1. Hides quality problems
 2. Hides inefficiencies
 3. Increases carrying costs
-

12. Elements of JIT System

1. Flow Manufacturing
2. Multi-Process Handling
3. Kanban
4. Manpower Reduction
5. Visual Control
6. Leveling
7. Changeover
8. Quality Assurance
9. Standard Operations

10. Jidoka

11. Maintenance & Safety

13. Steps to Establish JIT

Step 1: Awareness Revolution

Management commitment and culture change.

Step 2: Implement 5S

Step 3: Flow Manufacturing

Step 4: Leveling (Heijunka)

Step 5: Standard Operations

14. Flow Manufacturing Requirements

Seven Requirements:

1. Continuous flow production
 2. Compact equipment
 3. One-piece flow
 4. Cycle time control
 5. Multi-process operations
 6. Multi-skilled workers
 7. Standing operations
-

15. Mass Production vs Lean Production

Feature	Mass Production	Lean Production
Founder	Henry Ford	Taiichi Ohno
Volume	High Volume	Low Volume / High Mix
Workforce	Low-skilled	Multi-skilled
Focus	Quantity	Quality & Flexibility

Feature	Mass Production	Lean Production
Inventory	High	Low
Defects	Accepted	Stop the line
Pricing	Cost-Plus	Market-Based
Flexibility	Low	High

Most Important Exam Questions

2 Marks

1. Define Lean Manufacturing.
2. What is TPS?
3. What is Kaizen?
4. What is Kanban?
5. Define Jidoka.
6. Define Poka-Yoke.
7. What are the 5S principles?

5 Marks

1. Explain the Five Principles of Lean.
2. Explain TIM WOOD wastes.
3. Explain House of Lean.
4. Explain 5S methodology.
5. Explain Kanban system.

10 Marks

1. Explain Toyota Production System (TPS).
 2. Explain Lean Manufacturing and its benefits.
 3. Explain Lean tools (5S, VSM, SOP, Cellular Layout, Kaizen).
 4. Explain JIT Production System and its elements.
 5. Compare Mass Production and Lean Production.
-

One-Page Revision Sheet

Lean = Eliminate Waste + Maximize Customer Value

Three Mu's

- Muda = Waste
- Mura = Variation
- Muri = Overburden

TIM WOOD

Transportation, Inventory, Motion, Waiting, Overprocessing, Overproduction, Defects

5S

Sort → Set in Order → Shine → Standardize → Sustain

TPS Pillars

- JIT
- Jidoka

Major Lean Tools

- 5S
- VSM
- SOP
- Cellular Layout
- Kaizen
- Kanban
- Jidoka
- Poka-Yoke

JIT Goal

Right Product + Right Quantity + Right Time

Kaizen

Continuous Improvement

Poka-Yoke

Mistake Proofing

Kanban

Visual Pull System

Jidoka

Automation with Human Intelligence

Exam Focus Areas

- ★ 5S Principles
- ★ House of Lean
- ★ TIM WOOD Wastes
- ★ Kanban Functions & Rules
- ★ Jidoka Principles
- ★ Poka-Yoke Types
- ★ JIT Elements & Steps
- ★ Mass vs Lean Production Comparison

WEEK 12 – MAINTENANCE & SUSTAINABLE OPERATIONS

MBA Operations Management – Exam Ready Summary

This module covers two major areas:

1. **Maintenance Management** – keeping equipment and facilities in efficient working condition.
 2. **Sustainable Operations** – balancing economic growth, environmental protection, and social responsibility through sustainable business practices.
-

PART A – MAINTENANCE MANAGEMENT

1. Maintenance – Meaning

Maintenance refers to:

Activities designed to keep resources in good working condition or restore them to operating status.

Maintenance helps:

- Increase operational efficiency
- Reduce operating costs
- Improve reliability
- Increase production capacity
- Improve safety
- Extend equipment life

Objectives of Plant Maintenance

1. Increase functional reliability.
 2. Achieve product quality through properly maintained equipment.
 3. Maximize equipment life.
 4. Minimize maintenance and operating costs.
 5. Maximize production capacity.
 6. Enhance worker safety.
-

2. Types of Maintenance

A. Preventive Maintenance (PM)

Definition

Periodic inspection and servicing intended to prevent failures before they occur.

Activities

- Inspection
- Cleaning
- Lubrication
- Minor repairs
- Scheduled servicing

Principle

"Prevention is better than cure."

Benefits

- Reduced breakdowns
 - Reduced downtime
 - Increased reliability
 - Longer equipment life
-

B. Breakdown Maintenance (BM)**Definition**

Repair work performed after equipment failure.

Characteristics

- Emergency based
- Unplanned
- Higher downtime
- Higher repair costs

Suitable When

- Equipment is simple
 - Few machines exist
 - Failure does not cause major losses
-

3. Other Maintenance Types

Planned Maintenance

Maintenance performed according to recorded procedures and schedules.

Corrective Maintenance

Actions taken to restore failed equipment and eliminate root causes.

Opportunistic Maintenance

Replacing other worn parts while equipment is already shut down.

Routine Maintenance

Regular activities such as:

- Cleaning
- Lubrication
- Inspection
- Minor adjustments

Predictive Maintenance

Predicting failures before they occur using monitoring techniques.

Condition-Based Maintenance (CBM)

Continuous monitoring of equipment condition using sensors.

Design-Out Maintenance

Eliminating maintenance problems through better design.

4. Preventive Maintenance System

Components

- Equipment identification
- Documentation
- Regular inspections
- Cleaning and lubrication
- Minor repairs
- Overhauls
- Failure analysis

Types of PM Schedule

1. Fixed Time Maintenance

Performed at predetermined intervals.

2. Condition-Based Maintenance

Performed based on actual equipment condition.

5. Preventive vs Breakdown Maintenance

Preventive Maintenance	Breakdown Maintenance
Planned	Unplanned
Prevents failures	Acts after failure
Lower overall cost	Higher repair cost
Less downtime	High downtime
Requires inspection	Emergency repairs
Improves reliability	Disrupts production

Exam Question: Compare Preventive Maintenance and Breakdown Maintenance.

6. Condition-Based Maintenance (CBM)

Definition

Maintenance based on actual machine condition rather than fixed schedules.

Monitoring Techniques

- Visual inspection
- Temperature monitoring
- Vibration analysis
- Lubricant analysis
- Leakage monitoring
- Noise monitoring

Objectives

1. Detect failures before occurrence.
2. Perform maintenance only when needed.
3. Reduce downtime.
4. Reduce maintenance cost.

Benefits

- ✓ Increased availability
 - ✓ Improved safety
 - ✓ Better quality
 - ✓ Better maintenance planning
-

7. Maintenance Planning & Scheduling

Job Planning

Answers:

What work?

How work?

Steps

1. Understand equipment.
 2. Develop repair plan.
 3. Prepare material list.
 4. Arrange tools and equipment.
 5. Estimate time.
 6. Provide safety instructions.
-

Job Scheduling

Answers:

Who will do it?

When will it be done?

Scheduling Factors

- Availability of manpower
- Availability of tools
- Availability of spares
- Past maintenance records

Tools Used

- Bar Charts
 - Network Techniques
-

8. Systematic Maintenance Elements

1. Codification & cataloguing
 2. Equipment history cards
 3. Operating manuals
 4. SOPs
 5. Maintenance Time Standards (MTS)
 6. Work permits
 7. Job cards
 8. Monitoring and feedback
 9. Maintenance documentation
-

9. Basic Maintenance Decisions

Management must decide:

- Centralized vs Decentralized maintenance
 - Contract vs In-house maintenance
 - Repair vs Replacement
 - Individual vs Group replacement
 - Standby equipment requirements
 - Maintenance capacity
 - Preventive vs Predictive maintenance
-

PART B – TOTAL PRODUCTIVE MAINTENANCE (TPM)

10. Total Productive Maintenance (TPM)

Definition

TPM is:

A management system for maximizing equipment productivity through participation of all employees.

Goal

Increase production while simultaneously improving:

- Equipment reliability
 - Employee morale
 - Job satisfaction
-

11. Meaning of TPM

T – Total

- Participation of all employees
- All departments involved

P – Productive

- Maximize efficiency
- Eliminate losses
- Zero accidents

M – Maintenance

- Improve equipment effectiveness
 - Focus on entire equipment lifecycle
-

12. TPM Philosophy

TPM aims for:

Zero Breakdown

Zero Defects

Zero Accidents

and maximum equipment effectiveness.

It promotes:

- Teamwork
 - Small group activities
 - Employee ownership
 - Continuous improvement
-

13. Why TPM is Important?

1. Dramatic Results

- Reduced breakdowns
- Reduced idle time
- Fewer defects
- Higher productivity
- Lower inventory
- Lower costs

2. Better Workplace

- Cleaner workplace
- Safer environment
- Better customer confidence

3. Improved Skills

- Worker involvement
 - Knowledge sharing
 - Continuous learning
-

14. TPM Objectives

1. Increase production.
2. Increase employee morale.
3. Minimize emergency maintenance.
4. Achieve:

- Zero Defects
 - Zero Breakdown
 - Zero Accidents
5. Involve employees at all levels.
 6. Create self-maintenance teams.
 7. Fulfil regulatory compliance.

15. TPM Targets

Production

- 80% Overall Production Efficiency
- 90% Overall Equipment Effectiveness (OEE)

Quality

- No customer complaints

Cost

- Minimize manufacturing cost

Delivery

- 100% delivery performance

Safety

- Accident-free workplace

Workforce

- Multi-skilled employees
-

16. TQM vs TPM

TQM	TPM
Focuses on Quality	Focuses on Equipment
Output-oriented	Input-oriented
Software-oriented	Hardware-oriented

TQM	TPM
Quality improvement	Elimination of losses

Similarity

Both require:

- Top management commitment
- Employee participation

PART C – SUSTAINABLE OPERATIONS

17. Sustainable Development

Definition (WCED, 1987)

"Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs."

18. Sustainability in Manufacturing

Goals:

- Maximize profit
- Minimize environmental impact
- Conserve resources
- Conserve energy
- Ensure employee and community safety

19. Weak vs Strong Sustainability

Weak Sustainability	Strong Sustainability
Focus on economic capital	Focus on natural capital
Resources can be substituted	Natural resources must be preserved
Economic growth priority	Environmental protection priority

20. Anthropocentric vs Non-Anthropocentric Views

Anthropocentric

- Human-centered view.
- Nature exists for human benefit.

Non-Anthropocentric

- Humans are not the center.
 - Nature has intrinsic value.
-

21. Triple Bottom Line (TBL)

TBL measures business success using:

People

Social responsibility.

Planet

Environmental responsibility.

Profit

Economic performance.

Formula

TBL = People + Planet + Profit (3Ps)

22. Triple Bottom Line Sustainability

Social Sustainability

Focus on:

- Community welfare
- Employee welfare
- CSR

Environmental Sustainability

Focus on:

- Pollution reduction
- Resource conservation

Economic Sustainability

Focus on:

- Profitability
 - Growth
 - Financial performance
-

23. Examples of TBL

Unilever

People

- Dove Self-Esteem Project
- Anti-discrimination initiatives

Society

- Lifebuoy Handwashing Campaign
-

DHL

Planet

- Go Green Solutions
 - Sustainable logistics
 - Lower emissions
-

24. Benefits of Triple Bottom Line

- ✓ Better CSR
 - ✓ Better public image
 - ✓ Environmental awareness
 - ✓ Attracts investors
 - ✓ Attracts customers
 - ✓ Long-term sustainability
-

25. Challenges of TBL

- Difficult to measure social impact
 - Difficult to measure environmental impact
 - Higher implementation costs
 - Greenwashing risks
 - Conflicting objectives
-

26. Corporate Social Responsibility (CSR)

CSR focuses on:

- Ethical business practices
- Environmental responsibility
- Social responsibility

Goal

Do what is right for all stakeholders.

27. Carbon Footprint

Definition

Total greenhouse gases generated by activities, products, or organizations.

Measured in:

- Tons of CO₂ equivalent emissions

Objective

Reduce:

- Carbon emissions
 - Greenhouse gases (GHG)
-

28. Life Cycle Assessment (LCA)

Evaluates environmental impact from:

Raw Material

→ Manufacturing

→ Transportation

- Use
- Disposal

Goal

Reduce environmental impact throughout the product lifecycle.

29. The 3Rs of Sustainability

Reduce

Use fewer resources.

Reuse

Use products again.

Recycle

Convert waste into reusable materials.

Remember: Reduce → Reuse → Recycle

30. Design for Sustainability

Product Design Considerations

- Material selection
- Packaging
- Transportation
- End-of-life disposal

Design for Disassembly

Products should be easy to:

- Dismantle
 - Recycle
 - Reuse
-

31. ISO 14000 Standards

Purpose

Environmental Management Standards.

Covers

- Environmental management
- Auditing
- Performance evaluation
- Life-cycle assessment

Benefits

- Reduced resource usage
- Lower energy consumption
- Reduced waste
- Better corporate image
- Better process efficiency

ISO 14001

Specifically addresses Environmental Management Systems (EMS).

 Expected Exam Questions**2 Marks**

1. Define maintenance.
 2. Define preventive maintenance.
 3. Define breakdown maintenance.
 4. What is TPM?
 5. What is sustainability?
 6. What is TBL?
 7. What is CSR?
 8. Define carbon footprint.
-

5 Marks

1. Explain types of maintenance.
2. Explain preventive maintenance system.
3. Explain condition-based maintenance.

4. Explain TPM objectives.
 5. Explain Triple Bottom Line approach.
 6. Explain Life Cycle Assessment.
-

10 Marks

1. Discuss TPM and its objectives.
 2. Compare preventive and breakdown maintenance.
 3. Explain TPM benefits and targets.
 4. Explain sustainable development and sustainability measures.
 5. Explain Triple Bottom Line with examples.
 6. Discuss ISO 14000 and environmental sustainability.
-



One-Page Quick Revision

Maintenance Types

- Preventive
- Breakdown
- Corrective
- Predictive
- Condition Based
- Routine
- Opportunistic

TPM Goals

- Zero Breakdown
- Zero Defects
- Zero Accidents

TPM Targets

- OPE $\geq$ 80%
- OEE $\geq$ 90%

Sustainability

Meeting present needs without compromising future generations.

Triple Bottom Line

 People

 Planet

 Profit

3Rs

Reduce → Reuse → Recycle

CSR

Ethical + Social + Environmental Responsibility

Carbon Footprint

Total greenhouse gas emissions.

ISO Standards

- ISO 14000 → Environmental Management
- ISO 14001 → Environmental Management System (EMS)

High Probability Exam Topics

- ★ Preventive vs Breakdown Maintenance
- ★ Condition-Based Maintenance (CBM)
- ★ TPM Objectives & Benefits
- ★ TPM vs TQM
- ★ Triple Bottom Line (3Ps)
- ★ Sustainability Concepts
- ★ CSR & Carbon Footprint
- ★ ISO 14000 / ISO 14001 Standards